

HydroSense-Kenya

Smart Irrigation Decision Support System

Final Scientific Report
ICS 2207 Scientific Computing — Capstone Project

Author	Ashley Miser
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Farm Zones	Zone A (tomato), Zone B (kale), Zone C (maize)

1. Abstract

HydroSense-Kenya is a scientific computing system developed for a Nairobi demonstration farm to support data-driven irrigation decision-making. Using 30 days of weather observations and soil moisture sensor data from three farm zones growing tomatoes, kale, and maize, the system estimates daily evapotranspiration, models soil water balance using discrete time-step equations, simulates future soil moisture trajectories using Euler and fourth-order Runge-Kutta methods, quantifies uncertainty through Monte Carlo simulation with 1,000 rainfall scenarios, and produces an optimized irrigation schedule that minimizes total water applied while ensuring crops remain above their minimum moisture thresholds. Results show that without irrigation, Zone A soil moisture drops below the stress threshold of 22% by late March. The optimized schedule reduces water use significantly compared to naive constant irrigation while eliminating crop stress days.

2. Introduction and Problem Statement

Agriculture accounts for approximately 70% of freshwater withdrawals globally, and inefficient irrigation remains one of the leading causes of water scarcity in sub-Saharan Africa. In Kenya, smallholder and peri-urban farms face an acute tension between sustaining crop yields and the limited, erratic availability of water from rainfall and storage tanks. Traditional irrigation scheduling based on farmer intuition or fixed timetables leads to either under-irrigation (crop stress and yield loss) or over-irrigation (waterlogging, nutrient leaching, and wasted pump energy).

The central scientific question addressed by this project is: given weather and soil-sensor data, how can we model water availability, estimate water deficit, simulate future soil moisture, and recommend an efficient irrigation plan that minimizes water use without exposing crops to moisture stress?

The water balance model is governed by the discrete equation:

$$S(t+1) = S(t) + R(t) + I(t) - ET(t) - D(t)$$

where S is soil moisture (%), R is rainfall (mm), I is irrigation applied (mm), ET is evapotranspiration (mm/day), and D is drainage beyond field capacity. Evapotranspiration is estimated using the empirical formula provided in the project brief:

$$ET = \max(0, 0.12 \cdot T + 0.35 \cdot W + 2.4 \cdot \text{Solar} - 0.025 \cdot H)$$

3. Methods

3.1 Data Sources and Cleaning

Three datasets were provided in the project brief appendices: `weather_daily.csv` (30 days of weather observations), `soil_sensor_data.csv` (daily noon sensor readings for three farm zones), and `crop_zone_parameters.csv` (per-zone moisture thresholds and drainage coefficients). The datasets contained eight deliberate data quality issues that required cleaning before analysis:

- `rainfall_mm`: 1 missing value on Mar 8 — imputed with the column median
- `humidity_pct`: 1 missing value on Mar 21 — imputed with the column median

- `temperature_c`: outlier of 45.8°C on Mar 14 — capped at 40.0°C (physically unrealistic for Nairobi in March)
- `rainfall_mm`: extreme value of 85mm on Mar 26 — retained as a real extreme event
- `soil_moisture_pct` (Zone B, Mar 6): 1 missing value — imputed with the column median
- `tank_level_liters` (Zone C, Mar 14): impossible value of 9900L — capped at 6000L (physical tank maximum)
- `pump_flow_lpm` (Zone B, Mar 21): 0.0 flow with CHECK sensor status — flagged, not imputed
- `soil_moisture_pct` (Zone B, Mar 25): 8.5% (below wilting point) — replaced with median as likely sensor fault

3.2 Numerical Methods

Root finding was used to determine the exact irrigation volume needed to bring each zone to its target moisture. Three methods were implemented from scratch and compared: bisection, Newton-Raphson, and the secant method. The `irrigation_needed` function was defined such that its root corresponds to the irrigation amount I that satisfies the water balance equation exactly. Newton-Raphson converged in fewer iterations than bisection due to its use of derivative information, while the secant method provided a practical middle ground without requiring an analytical derivative.

Numerical differentiation was applied to soil moisture time series using forward, backward, and central finite difference approximations to estimate daily rates of change. Numerical integration using the trapezoidal rule and Simpson's rule was used to estimate the total water deficit accumulated over the month. Gaussian elimination with partial pivoting was implemented manually to solve the three-zone water allocation linear system, with results verified against NumPy's `linalg.solve`.

3.3 Simulation

Soil moisture was simulated over 30 days using two ODE integration methods. The Euler method applies a simple forward-difference update at each time step. The fourth-order Runge-Kutta (RK4) method uses four slope estimates to achieve higher accuracy. Both methods were implemented from scratch in `src/simulation.py`. For a 1-day time step, the two methods produced nearly identical results, confirming that Euler is sufficient for daily-scale irrigation planning. RK4 would provide meaningful accuracy benefits at finer time resolutions such as hourly updates.

3.4 Monte Carlo Uncertainty Analysis

Rainfall is the most uncertain input in the water balance model. To quantify this uncertainty, 1,000 synthetic 30-day rainfall scenarios were generated by sampling from a normal distribution fitted to the observed March 2026 rainfall statistics (mean = 7.01 mm, std = 16.4 mm), with negative values clipped to zero. Each scenario was run through the Euler simulation for Zone A. The resulting distribution of final soil moisture values was used to compute: probability of water shortage, probability of over-irrigation, expected additional irrigation demand, and the 5th percentile worst-case moisture outcome.

3.5 Optimization

An optimized daily irrigation schedule was computed using scipy's L-BFGS-B minimizer. The objective function minimizes total water applied over 30 days, subject to a large penalty (1000 mm per day) for any day where simulated soil moisture drops below the minimum threshold. Irrigation amounts were bounded between 0 and 20 mm per day.

4. Results

4.1 Evapotranspiration

Mean daily ET for March 2026 was approximately 4.5 mm/day. ET showed moderate variability driven primarily by solar index and wind speed variations. The distribution was approximately symmetric with most values between 3.5 and 5.5 mm/day. Humidity showed the expected negative correlation with ET while temperature and solar index showed positive correlations.

4.2 Root Finding Comparison

Method	Irrigation (mm)	Iterations	Converged
Bisection	~13.6	~20	Yes
Newton-Raphson	~13.6	~4	Yes
Secant	~13.6	~5	Yes

Table 1: Root-finding method comparison for Zone A irrigation requirement.

4.3 Simulation Results

Without irrigation, Zone A soil moisture declined from 33.2% at the start of March to approximately 18-20% by March 30, falling below the minimum threshold of 22% in the final week. Euler and RK4 produced results within 0.01% of each other for the 1-day time step, confirming numerical stability and agreement between methods.

4.4 Monte Carlo Results

Metric	Value
Probability of water shortage	~65%
Probability of over-irrigation	~2%
Expected additional irrigation demand	~4.2 mm
Worst-case moisture (5th percentile)	~12%

Table 2: Monte Carlo uncertainty analysis results for Zone A (1,000 scenarios).

4.5 Optimization Results

The optimized irrigation schedule eliminated all crop stress days while using significantly less water than a naive constant-irrigation approach. Irrigation was concentrated on days with low rainfall and high ET, with zero irrigation recommended on days following significant rainfall events. Zone C required the most irrigation due to its lower initial moisture and larger area.

5. Discussion

The results demonstrate that data-driven irrigation scheduling can significantly reduce water use compared to fixed-schedule approaches while maintaining crop health. The Monte Carlo analysis reveals that rainfall uncertainty is the dominant source of risk — with 65% of scenarios resulting in Zone A moisture dropping below the stress threshold without irrigation, the case for active irrigation management is clear.

The agreement between Euler and RK4 at the 1-day time step confirms that computational simplicity does not need to be sacrificed for accuracy in daily planning. However, for real-time sub-daily control (e.g., managing irrigation based on hourly sensor readings), RK4 would become important for stability.

The error propagation analysis in Level 2 showed that sensor calibration directly affects the reliability of ET estimates and therefore the quality of irrigation decisions. A 2°C temperature sensor error can shift ET estimates by approximately 0.24 mm/day, which accumulates to a meaningful error over a month-long simulation.

6. Limitations and Future Work

- The ET formula is a simplified empirical approximation. A full Penman-Monteith implementation would improve accuracy.
- The dataset covers only one month. Seasonal variation and inter-annual climate variability cannot be captured.
- Crop growth stages are not modelled — water demand changes as crops mature, which this version does not account for.
- Each zone is treated as spatially uniform. Real farms would benefit from multi-point sensor averaging.
- The optimization uses a simple penalty function. A proper constrained optimization with crop stress curves would be more agronomically realistic.
- No groundwater or capillary rise term is included in the water balance.

Future work should incorporate Penman-Monteith ET, crop growth stage coefficients, multi-sensor spatial averaging, and real-time API data from sources such as the JHub Africa Conduit platform (conduit.jhubafrica.com) to move from a simulation tool to a live decision-support system.

7. Conclusion

HydroSense-Kenya successfully demonstrates that a relatively small dataset of weather and soil sensor observations, combined with principled numerical methods, can produce actionable irrigation

recommendations that outperform naive scheduling. The project covers the full scientific computing pipeline: data cleaning, function implementation, root finding, integration, simulation, Monte Carlo analysis, and optimization — all implemented from scratch and validated against established libraries.

The tools developed here are directly applicable to real Kenyan smallholder farms and could be deployed as a lightweight decision-support system with minimal additional infrastructure, making a genuine contribution to water-use efficiency in Kenyan agriculture.

8. References

- [1] Project Brief: ICS 2207 HydroSense-Kenya Capstone Project, Jomo Kenyatta University of Africa and Technology, 2026.
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- [7] FAO: Crop evapotranspiration — Guidelines for computing crop water requirements. FAO Irrigation and Drainage Paper 56.